Experiment 7

AIM: Use Selenium to create browser-based tests

Step 1 Import Required Libraries

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.chrome.service import Service
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
from webdriver_manager.chrome import ChromeDriverManager
import time
```

Step 2 Start Browser

```
service = Service(ChromeDriverManager().install())
driver = webdriver.Chrome(service=service)
driver.get("https://the-internet.herokuapp.com/login")
driver.maximize_window()
```

Step 3 Wait for Page Elements

```
wait = WebDriverWait(driver, 10)
```

Step 4 Enter Login Credentials

```
username = wait.until(
EC.presence_of_element_located((By.ID, "username"))
)
username.send_keys("tomsmith")
driver.find_element(By.ID, "password").send_keys("SuperSecretPassword!")
```

Step 5 Click Login Button

```
driver.find_element(By.CSS_SELECTOR, "button.radius").click()
```

Step 6 Verify Login Success

```
message = wait.until(
EC.presence_of_element_located((By.ID, "flash"))
)
print("Login Message:", message.text)
```

Step 7 Logout

```
driver.find_element(By.CSS_SELECTOR, "a.button.secondary.radius").click()
```

Step 8 Close Browser

```
time.sleep(3)
driver.quit()
```

Command Prompt:

```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\RAJNISH\OneDrive\Desktop\selenium-lab-096>python --version
Python 3.14.3

C:\Users\RAJNISH\OneDrive\Desktop\selenium-lab-096>pip install selenium
Collecting selenium
  Downloading selenium-4.41.0-py3-none-any.whl.metadata (7.5 kB)
Collecting certifi>=2026.1.4 (from selenium)
  Downloading certifi-2026.2.25-py3-none-any.whl.metadata (2.5 kB)
Collecting trio<1.0,>=0.31.0 (from selenium)
  Downloading trio-0.33.0-py3-none-any.whl.metadata (8.5 kB)
Collecting trio-websocket<1.0,>=0.12.2 (from selenium)
  Downloading trio_websocket-0.12.2-py3-none-any.whl.metadata (5.1 kB)
Collecting typing_extensions<5.0,>=4.15.0 (from selenium)
  Downloading typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)
Collecting urllib3<3.0,>=2.6.3 (from urllib3[socks]<3.0,>=2.6.3->selenium)
  Downloading urllib3-2.6.3-py3-none-any.whl.metadata (6.9 kB)
Collecting websocket-client<2.0,>=1.8.0 (from selenium)
  Downloading websocket_client-1.9.0-py3-none-any.whl.metadata (8.3 kB)
Collecting attrs>=23.2.0 (from trio<1.0,>=0.31.0->selenium)
  Downloading attrs-26.1.0-py3-none-any.whl.metadata (8.8 kB)
Collecting sortedcontainers (from trio<1.0,>=0.31.0->selenium)
  Downloading sortedcontainers-2.4.0-py2.py3-none-any.whl.metadata (10 kB)
Collecting idna (from trio<1.0,>=0.31.0->selenium)
  Downloading idna-3.11-py3-none-any.whl.metadata (8.4 kB)
Collecting outcome (from trio<1.0,>=0.31.0->selenium)
  Downloading outcome-1.3.0.post0-py2.py3-none-any.whl.metadata (2.6 kB)
Collecting sniffio>=1.3.0 (from trio<1.0,>=0.31.0->selenium)
  Downloading sniffio-1.3.1-py3-none-any.whl.metadata (3.9 kB)
Collecting cffi>=1.14 (from trio<1.0,>=0.31.0->selenium)
  Downloading cffi-2.0.0-cp314-cp314-win_amd64.whl.metadata (2.6 kB)
Collecting wsproto>=0.14 (from trio-websocket<1.0,>=0.12.2->selenium)
```

```
  Downloading wsproto-1.3.2-py3-none-any.whl.metadata (5.2 kB)
Collecting pysocks!=1.5.7,<2.0,>=1.5.6 (from urllib3[socks]<3.0,>=2.6.3->selenium)
  Downloading PySocks-1.7.1-py3-none-any.whl.metadata (13 kB)
Collecting pycparser (from cffi>=1.14->trio<1.0,>=0.31.0->selenium)
  Downloading pycparser-3.0-py3-none-any.whl.metadata (8.2 kB)
Collecting h11<1,>=0.16.0 (from wsproto>=0.14->trio-websocket<1.0,>=0.12.2->selenium)
  Downloading h11-0.16.0-py3-none-any.whl.metadata (8.3 kB)
Downloading selenium-4.41.0-py3-none-any.whl (9.5 MB)
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776.6 kB/s  0:00:12
Downloading trio-0.33.0-py3-none-any.whl (510 kB)
Downloading trio_websocket-0.12.2-py3-none-any.whl (21 kB)
Downloading typing_extensions-4.15.0-py3-none-any.whl (44 kB)
Downloading urllib3-2.6.3-py3-none-any.whl (131 kB)
Downloading PySocks-1.7.1-py3-none-any.whl (16 kB)
Downloading websocket_client-1.9.0-py3-none-any.whl (82 kB)
Downloading attrs-26.1.0-py3-none-any.whl (67 kB)
Downloading certifi-2026.2.25-py3-none-any.whl (153 kB)
Downloading cffi-2.0.0-cp314-cp314-win_amd64.whl (185 kB)
Downloading outcome-1.3.0.post0-py2.py3-none-any.whl (10 kB)
Downloading sniffio-1.3.1-py3-none-any.whl (10 kB)
Downloading wsproto-1.3.2-py3-none-any.whl (24 kB)
Downloading h11-0.16.0-py3-none-any.whl (37 kB)
Downloading idna-3.11-py3-none-any.whl (71 kB)
Downloading pycparser-3.0-py3-none-any.whl (48 kB)
Downloading sortedcontainers-2.4.0-py2.py3-none-any.whl (29 kB)
Installing collected packages: sortedcontainers, websocket-client, urllib3, typing_extensions, sniffio,
pysocks, pycparser, idna, h11, certifi, attrs, wsproto, outcome, cffi, trio, trio-websocket, selenium
Successfully installed attrs-26.1.0 certifi-2026.2.25 cffi-2.0.0 h11-0.16.0 idna-3.11 outcome-1.3.0.post0
pycparser-3.0 pysocks-1.7.1 selenium-4.41.0 sniffio-1.3.1 sortedcontainers-2.4.0 trio-0.33.0 trio-
websocket-0.12.2 typing_extensions-4.15.0 urllib3-2.6.3 websocket-client-1.9.0 wsproto-1.3.2

[notice] A new release of pip is available: 25.3 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip

C:\Users\RAJNISH\OneDrive\Desktop\selenium-lab-096>pip install webdriver-manager
Collecting webdriver-manager
  Downloading webdriver_manager-4.0.2-py2.py3-none-any.whl.metadata (12 kB)
Collecting requests (from webdriver-manager)
  Downloading requests-2.32.5-py3-none-any.whl.metadata (4.9 kB)
Collecting python-dotenv (from webdriver-manager)
  Downloading python_dotenv-1.2.2-py3-none-any.whl.metadata (27 kB)
```
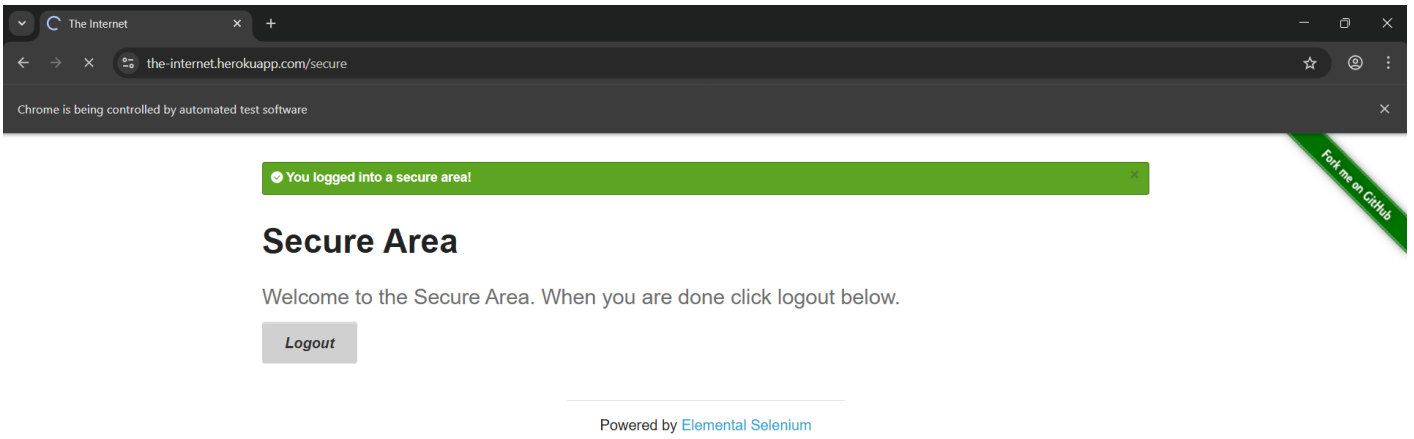

```
Collecting packaging (from webdriver-manager)
  Downloading packaging-26.0-py3-none-any.whl.metadata (3.3 kB)
Collecting charset_normalizer<4,>=2 (from requests->webdriver-manager)
  Downloading charset_normalizer-3.4.6-cp314-cp314-win_amd64.whl.metadata (41 kB)
Requirement already satisfied: idna<4,>=2.5 in
c:\users\RAJNISH\appdata\local\programs\python\python314\lib\site-packages (from requests-
>webdriver-manager) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in
c:\users\RAJNISH\appdata\local\programs\python\python314\lib\site-packages (from requests-
>webdriver-manager) (2.6.3)
Requirement already satisfied: certifi>=2017.4.17 in
c:\users\RAJNISH\appdata\local\programs\python\python314\lib\site-packages (from requests-
>webdriver-manager) (2026.2.25)
Downloading webdriver_manager-4.0.2-py2.py3-none-any.whl (27 kB)
Downloading packaging-26.0-py3-none-any.whl (74 kB)
Downloading python_dotenv-1.2.2-py3-none-any.whl (22 kB)
Downloading requests-2.32.5-py3-none-any.whl (64 kB)
Downloading charset_normalizer-3.4.6-cp314-cp314-win_amd64.whl (154 kB)
Installing collected packages: python-dotenv, packaging, charset_normalizer, requests, webdriver-
manager
Successfully installed charset_normalizer-3.4.6 packaging-26.0 python-dotenv-1.2.2 requests-2.32.5
webdriver-manager-4.0.2

[notice] A new release of pip is available: 25.3 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```



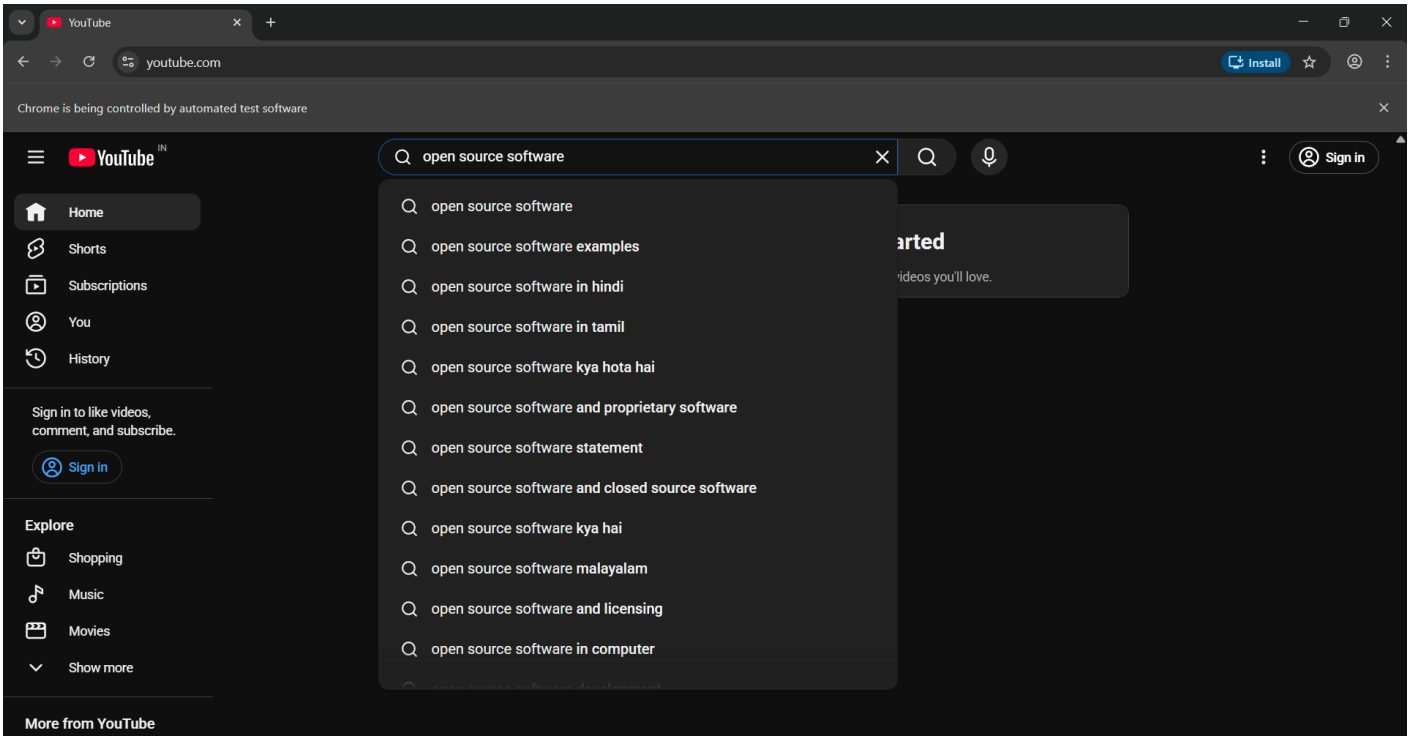

```
C:\Users\RAJNISH\OneDrive\Desktop\selenium-lab-096>python selenium_login_test.py
Login Message: You logged into a secure area!

C:\Users\RAJNISH\OneDrive\Desktop\selenium-lab-096>
```

Task: Automate YouTube Search using Selenium:

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.chrome.service import Service
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
from webdriver_manager.chrome import ChromeDriverManager
import time

# Start browser
driver = webdriver.Chrome(service=Service(ChromeDriverManager().install()))

# Open YouTube
driver.get("https://www.youtube.com")
driver.maximize_window()

# Wait object
wait = WebDriverWait(driver, 10)

# Find search box
search = wait.until(
    EC.presence_of_element_located((By.NAME, "search_query"))
)

search.send_keys("open source software")

# Click search button
search_btn = wait.until(
    EC.element_to_be_clickable((By.XPATH, '//*[@id="search-icon-legacy"]'))
)

search_btn.click()

# Wait to see results
time.sleep(5)

driver.quit()
```

Command Prompt:

```
C:\Users\RAJNISH\OneDrive\Desktop\selenium-lab-096>python selenium_login_test.py
Traceback (most recent call last):
  File "C:\Users\RAJNISH\OneDrive\Desktop\selenium-lab-096\selenium_login_test.py", line 27, in
<module>
    search_btn = wait.until(
        EC.element_to_be_clickable((By.XPATH, '//*[@id="search-icon-legacy"]'))
    )
  File "C:\Users\RAJNISH\AppData\Local\Programs\Python\Python314\Lib\site-
packages\selenium\webdriver\support\wait.py", line 121, in until
    raise TimeoutException(message, screen, stacktrace)
selenium.common.exceptions.TimeoutException: Message:
Stacktrace:
        chromedriver!GetHandleVerifier [0xc5cdf3+10b03]
        chromedriver!GetHandleVerifier [0xc5cf24+10c34]
        chromedriver!(No symbol) [0xa42120]
        chromedriver!(No symbol) [0xa8abca]
        chromedriver!(No symbol) [0xa8ae6b]
        chromedriver!(No symbol) [0xacd0b2]
        chromedriver!(No symbol) [0xaadb54]
        chromedriver!(No symbol) [0xaca9a5]
        chromedriver!(No symbol) [0xaad8a6]
        chromedriver!(No symbol) [0xa80229]
        chromedriver!(No symbol) [0xa80fe4]
        chromedriver!GetHandleVerifier [0xec48b9+2785c9]
        chromedriver!GetHandleVerifier [0xebfeb5+273bc5]
        chromedriver!GetHandleVerifier [0xede06b+291d7b]
        chromedriver!GetHandleVerifier [0xc75cc8+299d8]
        chromedriver!GetHandleVerifier [0xc7d9fd+3170d]
        chromedriver!GetHandleVerifier [0xc658c8+195d8]
        chromedriver!GetHandleVerifier [0xc65a92+197a2]
        chromedriver!GetHandleVerifier [0xc4ee9a+2baa]
        KERNEL32!BaseThreadInitThunk [0x76335d49+19]
        ntdll!RtlInitializeExceptionChain [0x77add81b+6b]
        ntdll!RtlGetAppContainerNamedObjectPath [0x77add7a1+231]
```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open source software examples
open source software in hindi
open source software in tamil
open source software kya hota hai
open source software and proprietary software
open source software statement
open source software and closed source software
open source software kya hai
open source software malayalam
open source software and licensing
open source software in computer
arted
ideos you'll love.